

Probability calibration with limited artist data in music tagging

Retrospective and prospective evidence on regularization and numerical reproducibility

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Preprint v1 • 6 October 2026 • Not peer reviewed

Abstract

Post hoc calibration can worsen probability estimates when few labeled artists are available. This study combines a retrospective music-tagging benchmark with a prespecified new-artist evaluation. The historical benchmark contains 1,206 tracks from 469 artist IDs, four overlapping genre labels, three audio representations and two weighting settings across 20 overlapping splits with artist-disjoint roles. Seven probability treatments use calibration budgets of 24, 47 or 94 artists. Identity-centered ridge Platt has lower mean Brier loss than ordinary Platt in all 18 historical aggregate conditions, yet higher loss than raw outputs in all six unweighted partial-budget conditions. Historical numerical verification reproduces selection and fallback decisions across macOS, Linux and Windows, with a maximum probability discrepancy of 1.836×10^{-13} . The subsequent same-source evaluation retains 143 calibration tracks from 94 new artist IDs and 637 evaluation tracks from 400 other new artist IDs. All selected tracks survive processing, and the cohort passes the coverage checks. The prespecified primary comparison averages ridge-minus-Platt losses across 20 fixed unweighted MAEST heads, five draws and the 24/47-artist budgets. Its Brier difference is -0.007050 , with a paired evaluation-artist bootstrap 95% interval of $[-0.008361, -0.005748]$, conditional on the fixed heads and calibration pool. This supports the average advantage over ordinary Platt. It does not establish an advantage over simpler controls: ridge point estimates remain worse than raw at both partial budgets, while analytical correction outperforms ridge in the weighted partial-budget comparisons. At 94 artists, ridge-minus-Platt intervals span zero under both weightings. The findings support retaining raw and analytical baselines and separating statistical evidence from numerical reproducibility. They concern noisy source tags in one music collection and do not establish a new algorithm or cross-source generalization.

Keywords: probability calibration; music tagging; limited calibration data; regularization; artist grouping; numerical reproducibility

1 Introduction

A music classifier may assign a track a probability of 0.8 for a genre even when similarly scored tracks carry that label much less often. Calibration concerns this agreement between numerical confidence and observed outcomes. In multi-label tagging, a track may be both electronic and pop, so the four probabilities studied here are separate binary predictions rather than mutually exclusive class probabilities. A useful calibration analysis must therefore evaluate each label and state what its reference labels mean.

Established approaches include sigmoid fitting, temperature scaling, isotonic regression and beta calibration [1, 2, 3, 4]. Their flexibility creates a practical question when few labeled artists are available: does the fitted adjustment correct a systematic distortion, or fit incidental features of the calibration sample? The raw score is a meaningful baseline, especially when the original classifier is already reasonably calibrated. A regularizer that favors the identity map can reduce large adjustments, but selecting its strength from the same small calibration pool introduces another source of variability.

This study uses preserved scores from a music-classification project to examine two related issues. The first is the dependence of probability quality on calibration budget and artist selection. The second is whether numerical solver termination changes cross-validation choices even when the data and statistical objectives are fixed. The contribution is a documented empirical comparison with retained unfavorable results and an audited implementation. Identity-centered regularization, calibration maps and bounded optimization are existing ideas; no algorithmic novelty is claimed.

All tracks in the retrospective benchmark have been examined in earlier project experiments. Within each historical split, classifier fitting, calibration and evaluation use disjoint artist IDs, but that separation does not undo prior project exposure. A subsequent protocol specifies a new-artist evaluation after excluding documented project exposure. The two stages address different evidential questions: the historical benchmark describes previously observed data, while the prospective comparison tests a fixed hypothesis on newly selected artists from the same source.

2 Related work

Platt introduced sigmoid mappings for classifier outputs [1], and Zadrozny and Elkan studied transforming scores into probability estimates using isotonic regression [3]. Temperature scaling provides a low-dimensional recalibration baseline [2]. Beta calibration models a broader family of monotone probability transformations [4]. The present Platt implementation acts on logits and contains the identity map. The inability of a logistic family acting directly on probabilities to reproduce identity must not be transferred to this implementation.

Regularization and internal cross-validation are established tools in calibration [5]. A recent preprint also explicitly penalizes deviations from an identity transformation [6]. These multiclass settings are not equivalent to the present binary implementation, but they establish prior use of the general shrinkage idea. Numerical stability and convergence in sigmoid fitting have likewise received specific attention [7].

Class weighting can itself distort a probability estimate. An analytical correction based on the fitting-sample class counts is therefore included, following the distinction between weighted training objectives and unweighted event probabilities [8]. Audio-specific work has also examined uncertainty calibration [9]. The current study differs in its narrow empirical design: the classifiers remain fixed, the number and identity of calibration artists vary, and both statistical and numerical sources of variation are retained.

The Brier score is a proper score for probabilistic forecasts [10]. It measures overall probability quality rather than calibration alone; reliability and resolution both affect it [11]. Calibration assessment also depends on the evaluation procedure and available sample size [12]. Accordingly, the historical analysis uses paired Brier differences as its primary descriptive outcome, retains log loss and label-level diagnostics, and avoids interpreting a lower Brier score as proof of reliability in every probability range.

3 Data and experimental design

3.1 Historical score bundle

The input bundle derives from MTG-Jamendo [13]. It contains 1,206 development tracks and 469 artist IDs, with broad source-tag targets for electronic, pop, ambient and rock. Positive counts are 435, 261, 244 and 220, respectively; labels may overlap. A fixed tag ontology maps related tags to these categories. For example, synthpop maps to electronic and pop, while poprock maps to pop and rock. These targets are noisy and incomplete source annotations, not labels adjudicated by listeners for this study.

Three preserved representations are compared: Discogs-MAEST and two MERT variants [14, 15]. The audited MAEST feature vectors have 2,304 dimensions. MERT-v0 and MERT-v1 use 768-dimensional means of saved layers 1 through 12. The source pipeline used the first 30 seconds of each recording; the retrospective analysis neither re-extracts audio features nor changes encoder weights. Each source case retains the fitted classifier scores and the identities of its fitting, calibration and evaluation tracks.

The 20 historical splits contain 281 fitting artists, 94 calibration artists and 94 evaluation artists each. Across splits, the corresponding track counts range from 710 to 736, 228 to 254 and 225 to 255. The source procedure selected splits from 256 artist-grouped candidates using size and label balance. Artist IDs are disjoint among roles within a split, but splits overlap with one another. The three representations use the same songs. They are not independent datasets. Source standardization and logistic heads were fitted using the fitting partition; their

regularization parameter was $C = 0.001$. The two weighting conditions are unweighted training and balanced weighting for ambient only. The other three heads provide exact paired controls.

Table 1. Scope and units of the experiment. Repeated executions are not additional independent songs or datasets.

Component	Specification
Tracks and artists	1,206 tracks; 469 artist IDs
Label positives	electronic 435; pop 261; ambient 244; rock 220
Source cases	$20 \text{ splits} \times 3 \text{ representations} \times 2 \text{ weightings} = 120$
Calibration budgets	24 / 47 / 94 artists
Calibration tracks	41–87 / 93–149 / 228–254 in the five follow-up draws
Follow-up stability analysis	1,800 executions; 1,320 distinct source/subset cases
Numerical reproduction including original draw	2,160 executions; 1,560 distinct source/subset cases

3.2 Calibration budgets and artist folds

For each outer split, a seeded artist permutation defines nested calibration subsets containing 24, 47 and 94 artists, corresponding to rounded-up budgets of 25%, 50% and 100%. Every track belonging to a selected artist is retained. Subsets are identical across representations and weighting settings for the same split and draw. There is no target balancing or redraw to obtain a favorable class composition. The original budget seed is 2026100101. Five follow-up seeds, 2026100201 through 2026100205, form the main stability analysis.

Ridge strength is selected independently for each label using three artist-grouped out-of-fold partitions of the selected calibration subset. The inner seed is 2026100102. An artist contributes all of its tracks to one fold. Changing the selected artists may also change their fold assignments, even though the assignment rule and inner seed are fixed. The repeated-draw analysis therefore measures variation under the complete subset-and-selection procedure; it does not isolate a pure effect of sample composition.

Each draw executes $20 \times 3 \times 2 \times 3 = 360$ source/budget cases. The five follow-up draws execute 1,800 cases but contain 1,320 distinct source/subset cases because their full-budget endpoints are identical. Including the separately reported original draw yields 2,160 executions and 1,560 distinct source/subset cases. The repeated full-budget endpoint is counted once per source case in scientific summaries.

3.3 Exposure and chronology

The first budget protocol was established on 1 October 2026 after earlier calibration work had been observed. The stability extension was specified on 2 October after the first budget results were known. A numerical portability protocol was then frozen before the six complete v2 runs. These are versioned execution plans for exploratory analyses, not a preregistration or a fresh-data confirmation. Historical manuscript tables were derived from the saved v2 outputs without refitting models. The original draw is retained separately in Appendix B rather than added as a sixth follow-up draw.

3.4 Prespecified new-artist cohort

After the historical results were known, a prospective protocol was frozen on 3 October 2026 before new-cohort audio acquisition. It specifies an evaluation within MTG-Jamendo using artists absent from the audited project-exposure records. Source metadata was available during eligibility checks; no new classifier outcomes informed selection. This is an internally timestamped execution protocol, without a claim of public preregistration or absence from encoder pretraining.

The eligible frame contains 1,912 tracks from 553 artist IDs in archives 20 through 39. Eligibility requires official genre metadata, a duration of at least 30 seconds, an audio checksum and one of the permitted recording licenses. All genres remain eligible, including tracks outside the four focus labels. A fixed hash ordering selects 94 calibration artists and 400 evaluation artists, with at most two tracks per artist. The frozen selection contains 143 calibration tracks and 637 evaluation tracks. These are selected counts; processing attrition and final support are reported separately in Table 4. This label-independent sampling differs from the historical coverage-enriched cohort. Track-weighted results describe this artist-sampled cohort with a two-track cap, rather than uniform track sampling from the full catalog.

All 20 historical MAEST classifier seeds are retained in both weighting settings, giving 40 fixed scaler-and-head sets. No head is selected by its historical performance. Neither their parameters nor the encoder weights are retrained. New audio uses the same first-30-second MAEST feature procedure. Acquisition failures and detected exact audio duplicates remove rows or implicated artists under the frozen rules, without replacement. Nominal calibration budgets remain 24, 47 and 94 artists; every analysis records the surviving subset and its original fold assignments. Appendix C describes the acquisition, calibration and integrity rules.

4 Calibration methods and evaluation

4.1 Seven probability treatments

Let z be a frozen binary classifier logit and σ the logistic function. Raw probabilities are $\sigma(z)$. No treatment normalizes probabilities across the four labels. Parametric maps minimize unweighted mean binary negative log likelihood on the selected calibration tracks, with the stated penalty added for ridge Platt. Parametric fitting starts from the identity map. The treatment definitions are as follows.

Table 2. Probability treatments. All bounds, candidate penalties and tie rules are fixed across v1 and v2.

Treatment	Definition
Raw	$\sigma(z)$; no calibration data.
Prior offset	Add $\log(n_+/n_-)$ to weighted ambient z only; fitting-partition counts.
Temperature	$\sigma(az)$, $a \in [0.001, 100]$; mean binary NLL.
Platt	$\sigma(az + b)$; bounded parameters; NLL with unsmoothed binary targets.
Beta	$\sigma(a \log p - b \log(1 - p) + c)$; bounded monotone map.
Isotonic	Nondecreasing least-squares map; endpoint clipping outside its fitting range.
Ridge Platt	$\text{NLL} + \lambda[(a - 1)^2 + b^2]$; artist OOF Brier selects λ or exact identity.

For Platt and ridge Platt, the calibrated probability is

$$q = \sigma(az + b) \quad (1)$$

where a lies in $[0.001, 100]$ and b in $[-20, 20]$. This implementation uses unsmoothed binary targets rather than Platt’s original target-smoothing procedure. Ridge Platt minimizes

$$J_\lambda(a, b) = L(a, b) + \lambda[(a - 1)^2 + b^2] \quad (2)$$

where L is mean binary negative log likelihood. The penalty shrinks toward $a = 1$ and $b = 0$, not toward zero slope. Candidate penalties are 0, 0.001, 0.01, 0.1 and 1, together with an exact identity option that does not require fitting. Out-of-fold Brier selects the candidate separately for each label. Scores within 10^{-12} of the minimum are tied; identity is preferred, followed by the strongest finite penalty. A selected finite-penalty map is refitted on all selected calibration tracks. This rule offers no guarantee of improving evaluation loss.

For an ambient head trained with balanced class weights, the analytical prior-offset treatment adds $\log(n_+/n_-)$ to its logit, where counts come only from the classifier-fitting labels. It leaves unweighted cases and the other three labels exactly unchanged. This correction follows an idealized weighted-risk relationship; it need not reproduce the solution of a separately trained, regularized unweighted head. It requires no calibration labels.

Beta calibration uses $\sigma(a \log p - b \log(1 - p) + c)$, with $p = \sigma(z)$, a and b in $[0.001, 100]$ and c in $[-20, 20]$. Log terms are computed stably from z . Isotonic regression fits a nondecreasing least-squares map of z and clips predictions outside the observed fitting range to its endpoint values. These bounded and clipping choices are part of the evaluated implementations and should not be generalized to every implementation of the same named method.

4.2 Outcomes and aggregation

The common outcome for both stages is macro binary Brier score, averaging equally over the four labels and over tracks within each evaluation partition:

$$BS = \frac{1}{4n} \sum_{j=1}^4 \sum_{i=1}^n (q_{ij} - y_{ij})^2 \quad (3)$$

In the historical analysis, the paired contrast for each treatment is $\Delta BS = BS(\text{treatment}) - BS(\text{raw})$; negative values favor the treatment. Ridge versus ordinary Platt is a second planned descriptive contrast. In the follow-up analysis, five draw-level differences are first averaged within each outer split, and the 20 outer-split means receive equal weight. Full-budget endpoints use one copy per split. Log loss clips probabilities to $[10^{-15}, 1 - 10^{-15}]$ only for scoring. Label-level Brier, average precision and fixed 5-bin and 10-bin reliability summaries remain available in the archived reports.

A direction reversal requires at least one improvement and one deterioration across the five draws, with an absolute tie tolerance of 10^{-12} . A tie alone is not a reversal. Ranges describe the observed draws and are not confidence intervals. Overlapping splits, repeated songs and unchanged control labels preclude treating all case executions as independent replicates. The historical analysis reports no split-level t-test, population confidence interval or estimate of an adequate minimum calibration sample.

4.3 Numerical verification

The original v1 implementation used L-BFGS-B. A small number of optimizer nonconvergences triggered declared identity fallbacks in inner folds, changing candidate scores across platforms. Version v2 replaces that numerical procedure with bounded Newton steps, analytic gradients and Hessians, compensated scalar summation and fixed-order Cholesky solves. Coordinate derivative-sign bisection is used when curvature or line search prevents progress. The statistical objectives, bounds, initialization, candidate grid and selection rule remain unchanged.

Success requires finite feasible parameters, no material increase in the objective, a box-projected gradient infinity norm at most 2×10^{-12} and a relative proposed parameter step at most 2×10^{-12} . The iteration limit is 300. Single-class calibration targets retain an explicitly logged identity fallback; numerical nonconvergence also remains a recorded failure state, but v2 acceptance requires zero numerical-failure fallbacks. Data corruption and programming errors stop execution rather than being converted into fallbacks.

An independent verification path reconstructs objectives, gradients, probabilities, out-of-fold selection and reported metrics. Cross-platform checks compare every evaluation probability and every candidate out-of-fold probability, including unselected candidates, at absolute tolerance 10^{-8} with zero relative tolerance. Selected penalties and all fallback decisions must match exactly. Numerical coefficients need not be unique in poorly identified designs; agreement of predictions and verified constrained stationarity is therefore more informative than identical iteration counts.

4.4 Prospective primary comparison and uncertainty

The new evaluation has one primary contrast: ridge Platt minus ordinary Platt under unweighted heads. For each evaluation track, the four-label squared-loss difference is averaged equally across all 20 fixed heads, five prespecified calibration draws and the nominal 24- and 47-artist budgets. These per-track differences are then averaged across evaluation tracks. The procedure averages losses, not probabilities, and does not form a prediction ensemble. Full-budget results, weighted heads and all other method comparisons are secondary.

Using the bootstrap principle [16], paired artist-cluster resampling uses 10,000 resamples of evaluation artists with replacement. Each sampled artist contributes all of its retained tracks, and each replicate divides summed loss differences by the sampled track count. The 2.5th and 97.5th percentiles give a two-sided 95% interval. The interval conditions on the fixed heads, calibration pool and subset plans. It does not include variation from acquiring another calibration population, retraining the classifiers, missingness or tag errors. Five subset draws within one pool are not independent replications. Secondary intervals use the same resamples and are unadjusted pointwise descriptions.

Primary support requires the prespecified coverage checks and an upper interval endpoint below zero. An interval spanning zero is inconclusive. Coverage requires at least 80% of selected calibration artists overall and in each nominal subset, at least 320 evaluation artists and 80% of selected evaluation tracks. Each label also requires at least 20 evaluation artists with a positive track and 20 whose retained tracks are all negative. A failed gate is reported as incomplete evidence regardless of the point estimate. The sample is not expanded or replaced after outcomes are observed.

5 Results

5.1 Budget dependence and the importance of raw controls

Figure 1 and Appendix A report all seven treatments in the historical benchmark. Ridge Platt has lower mean Brier score than ordinary Platt in all 18 representation-by-weighting-by-budget conditions. This advantage is conditional on the comparator. With unweighted classifiers, ridge Platt is worse than raw at both 24 and 47 artists for all three representations. For MAEST, mean raw Brier is 0.132160; at 24 artists the ridge score is 0.135346 and the ordinary Platt score is 0.141133. Regularization reduces the degradation without removing it.

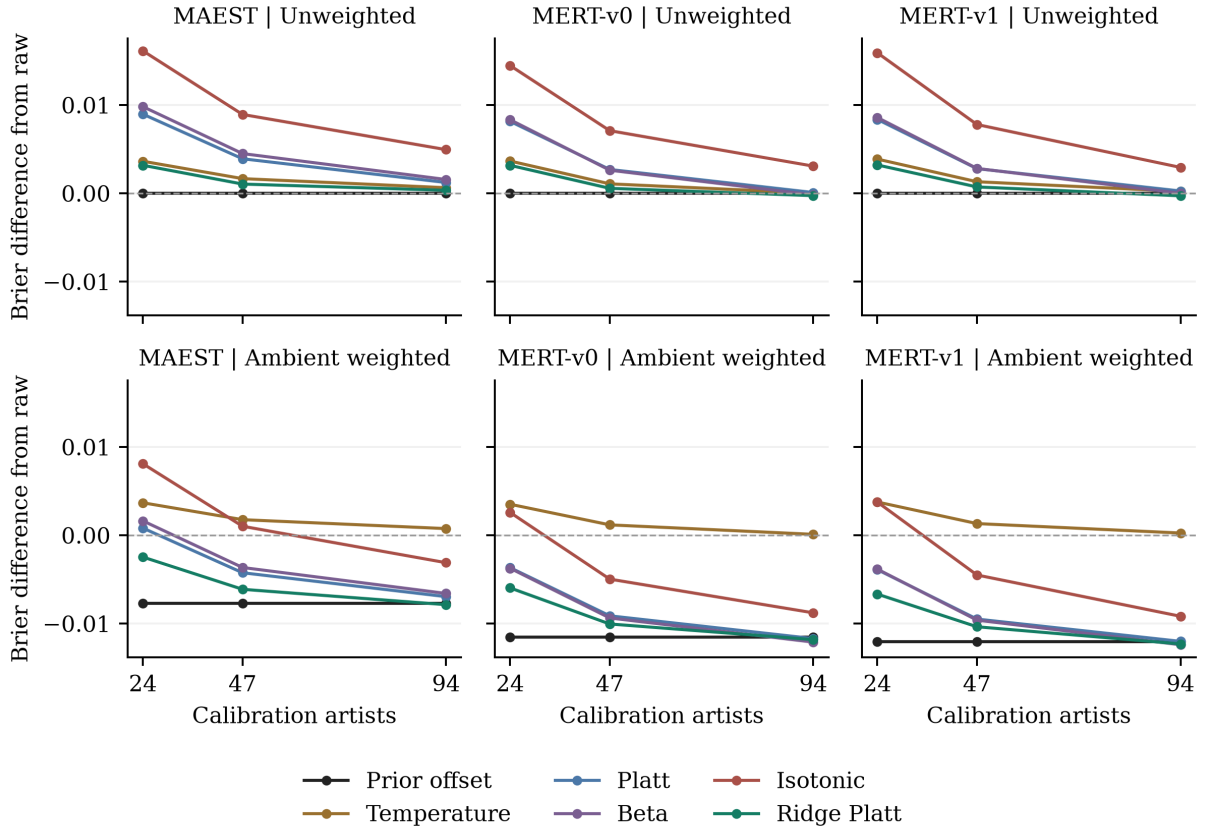


Figure 1. Mean paired Brier differences from raw across the five follow-up draws. Values below zero favor calibration. The 94-artist endpoint is counted once per outer split. Lines connect the three tested budgets and do not estimate intermediate budgets. Panels share the same vertical scale; no confidence intervals are implied.

The ambient-weighted setting gives calibrators a systematic distortion to correct. For MAEST, raw Brier is 0.140444 and ridge Platt reaches 0.137985, 0.134335 and 0.132595 as the budget grows. However, the calibration-free prior offset already yields 0.132749. Across all three representations, this analytical correction has lower mean Brier than ridge Platt at both partial budgets. A comparison restricted to fitted calibrators would therefore overstate the practical case for fitting a map from a small calibration pool.

Additional flexibility does not consistently help. In unweighted MAEST with 24 artists, beta and isotonic scores are 0.141982 and 0.148302, respectively, compared with raw 0.132160. At full budget, some MERT conditions favor beta or ridge by small margins, while unweighted MAEST still favors raw. These are descriptive comparisons on exposed data, not a rule for choosing a deployment method after inspecting evaluation scores.

5.2 Variation across selected artists

The average difference can conceal sensitivity to artist selection. In ambient-weighted MAEST with 24 artists, ridge Platt has mean $\Delta BS = -0.002458$, but the direction reverses in 15 of the 20 outer splits. The mean within-split range across five draws is 0.010933. At 47 artists, mean ΔBS is -0.006109 , reversals occur in 2 of 20 splits and the mean range is 0.004989. Figure 2 shows the individual split ranges rather than a pooled uncertainty band.

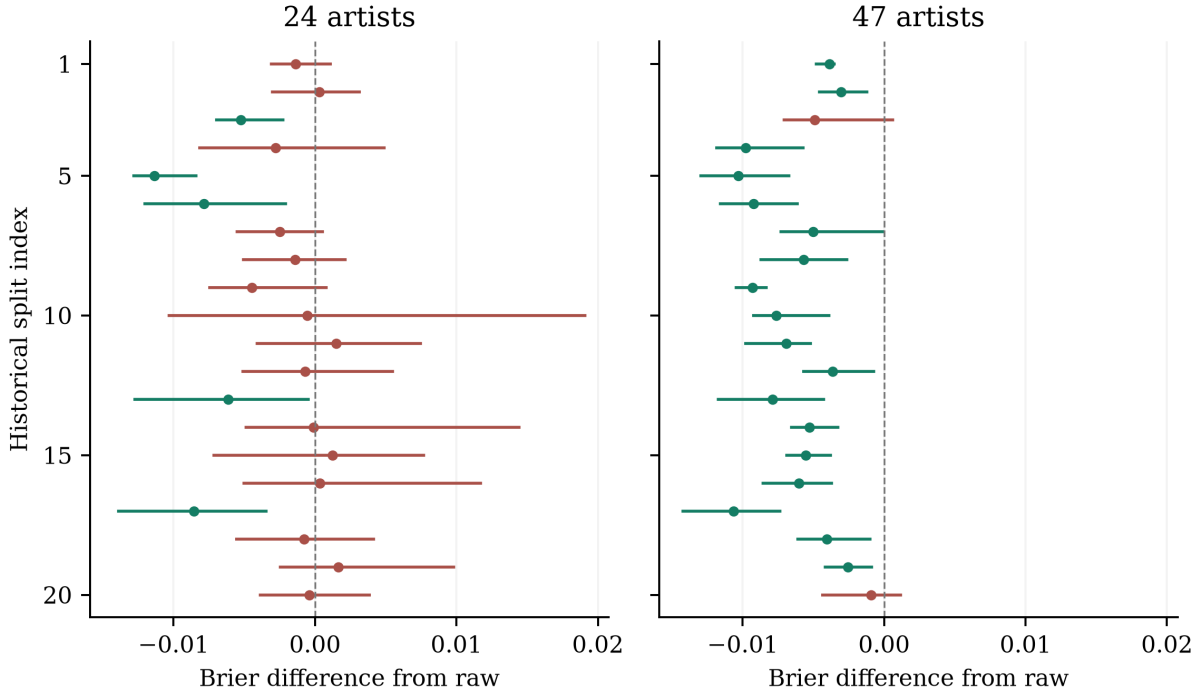


Figure 2. Sensitivity of ridge Platt versus raw for ambient-weighted MAEST. Each horizontal segment spans the smallest and largest difference across five draws for one historical split; the dot is their mean. The same split ordering is used in both panels. Segments are observed ranges, not confidence intervals.

For unweighted MAEST, mean differences are positive at both partial budgets: +0.003186 and +0.001061. Direction reversals occur in 7 and 13 of 20 splits, respectively. Across all representations, weightings, partial budgets and labels, the selected ridge option changes across draws in 870 of 960 design conditions. This count includes duplicated unaffected labels across weighting settings and is not an independent-sample frequency. It indicates sensitivity of the selection procedure, not that every change in penalty causes a large change in predicted probability.

5.3 Cross-platform agreement

All three tested systems complete the six frozen v2 plans and pass 166 tests each. On each system, all 159,327 optimizer fits converge; there are zero numerical-failure fallbacks and 360 declared single-class inner-fit fallbacks. The 600 duplicate full-budget executions match exactly within each environment. Selected ridge penalties and fallback flags, reasons and counts agree across macOS, Linux and Windows.

Table 3. Largest absolute probability differences against the CI macOS reference. The prespecified acceptance tolerance is 10^{-8} . Counts and selections are compared exactly.

Comparison	Evaluation probabilities	Candidate OOF probabilities
Linux / CI macOS	1.160738×10^{-13}	1.835754×10^{-13}
Windows / CI macOS	1.160738×10^{-13}	1.835754×10^{-13}
Local macOS / CI macOS	0	0

Each platform comparison covers 14,563,584 evaluation probabilities, 7,250,400 candidate out-of-fold probabilities, 51,840 candidate scores and 42 numeric report files. The largest probability difference is 1.835754×10^{-13} , and the largest candidate-score difference is 1.046385×10^{-14} . These values demonstrate agreement for the frozen inputs, software versions and tested environments. They are not a universal bitwise-determinism guarantee or an independent replication on new data.

5.4 A numerical correction can worsen evaluation performance

The v1-to-v2 comparison retains three changed label-level selections and 46 changed fallback locations. After duplicate full-budget cases are removed, mean ridge Brier increases by 3.520302×10^{-6} across the 1,560 source/subset cases. The largest individual probability change is 0.404575, so some changes are substantive model-selection changes rather than floating-point noise. Beta mean Brier decreases by 1.377194×10^{-6} , whereas raw, prior-offset and isotonic predictions remain unchanged.

One diagnostic case occurs for MERT-v0 rock at the 24-artist budget. An inner training fold has 42 tracks and four positive labels. More complete optimization drives the unpenalized fit toward its existing parameter bounds on a separable sample. The candidate out-of-fold Brier rises from 0.093787 to 0.143780, while the identity score is 0.094201. The selected option changes from zero penalty to identity. The resulting evaluation macro Brier increases by approximately 0.002722 in each paired weighting condition; those two conditions share the rock head and are not independent replications. The case separates numerical accuracy of optimization from generalization of the selected map.

5.5 Prespecified new-artist evaluation

All 780 selected tracks were retrieved and passed feature extraction, with no artist or track removed by the frozen duplicate rules (Table 4). The calibration pool contains 143 tracks from 94 artist IDs; evaluation contains 637 tracks from 400 different artist IDs. Positive evaluation artists number 157, 88, 68 and 99 for electronic, pop, ambient and rock, respectively. All coverage gates pass. The primary mean ridge-minus-Platt Brier difference is -0.007050 , with a conditional paired artist-bootstrap 95% interval of $[-0.008361, -0.005748]$ (Table 5). This supports the prespecified average comparison across the 20 unweighted heads, five draws and two partial budgets.

The advantage depends on the comparator. Unweighted raw Brier is 0.125033, compared with ridge values of 0.128756 and 0.125368 at 24 and 47 artists. Ridge-minus-raw differences are $+0.003723$ and $+0.000335$; the latter pointwise interval, $[-0.000009, +0.000673]$, includes zero. At 24 artists, ridge has higher Brier for all four labels, with the largest increase for pop, from 0.136982 to 0.144026. In the weighted condition, the calibration-free analytical offset reaches 0.124820, below ridge at both partial budgets, 0.131265 and 0.126142. No fitted treatment has lower mean Brier than raw in the unweighted partial-budget comparisons or than offset in their weighted counterparts (Table C1). These are secondary comparisons, not additional confirmatory findings.

The partial-budget result does not extend to a claim of ridge superiority at 94 artists. With unweighted heads, ridge Brier is 0.124756 and ordinary Platt is 0.124404; their difference is $+0.000352$, with a pointwise interval of $[-0.001064, +0.001758]$. The corresponding weighted ridge-minus-Platt interval also includes zero. Tables C1 and C2 retain all seven treatments and both scoring rules. In particular, lower average Brier relative to one fitted comparator should not be interpreted as reliable probabilities throughout the score range or as a rule for selecting a deployment method after examining evaluation outcomes.

The new run completed 32,508 recorded optimized fits without a numerical or single-class fallback. A separate reconstruction within this project checked 124,800 head logits, 7,847,840 probabilities and 1,760 ridge out-of-fold selections. Logits matched exactly; the maximum probability discrepancy was 1.665×10^{-16} , and the bootstrap reconstruction differed by at most 3.469×10^{-18} . The saved chronology places the protocol before acquisition and finalized predictions before evaluation-label parsing. This new numerical check was performed on macOS; it is separate from the historical three-system comparison and from replication by another research group.

Table 4. Selected and retained counts in the new-artist cohort. Nominal budgets identify the frozen subsets; actual artist counts reflect acquisition and duplicate exclusions.

Stage	Calibration artists / tracks	Evaluation artists / tracks
Selected	94 / 143	400 / 637
Downloaded	94 / 143	400 / 637
After duplicate exclusion	94 / 143	400 / 637
Final usable cohort	94 / 143	400 / 637

Table 5. Primary ridge-minus-Platt Brier contrast and prespecified secondary comparisons on the new cohort. Intervals resample evaluation artists and condition on the fixed heads and calibration pool. Secondary intervals are unadjusted pointwise descriptions.

Setting	Budget	Contrast	Difference and 95% interval
Primary U	24 + 47	Ridge – Platt	-0.007050 [-0.008361, -0.005748]
Secondary U	24	Ridge – Platt	-0.010800 [-0.012387, -0.009192]
Secondary U	24	Ridge – Raw	+0.003723 [+0.002879, +0.004572]
Secondary U	47	Ridge – Platt	-0.003300 [-0.004487, -0.002134]
Secondary U	47	Ridge – Raw	+0.000335 [-0.000009, +0.000673]
Secondary U	94	Ridge – Platt	+0.000352 [-0.001064, +0.001758]
Secondary U	94	Ridge – Raw	-0.000277 [-0.000825, +0.000268]
Secondary W	24	Ridge – Platt	-0.008246 [-0.009940, -0.006524]
Secondary W	24	Ridge – Raw	-0.005477 [-0.008899, -0.002016]
Secondary W	47	Ridge – Platt	-0.002594 [-0.003862, -0.001347]
Secondary W	47	Ridge – Raw	-0.010601 [-0.014039, -0.007184]
Secondary W	94	Ridge – Platt	+0.000130 [-0.001192, +0.001450]
Secondary W	94	Ridge – Raw	-0.012121 [-0.015861, -0.008323]

6 Discussion and limitations

Both stages support retaining a credible no-adjustment option and a mechanism-based baseline. Ridge regularization improves the comparison with ordinary Platt in all historical aggregate conditions and in the prespecified new-cohort partial-budget mean. That advantage does not establish an improvement over raw scores, and the new full-budget comparison does not show the same directional advantage. When ambient weighting creates a predictable offset, correction using fitting counts can outperform a map estimated from a small calibration sample. The practical choice therefore depends on the original probability distortion and on which simpler control is available.

The repeated artist draws also change the interpretation of a favorable average. A negative mean difference can coexist with reversals within most outer splits. Reporting the mean alone would obscure this dependence. Five draws provide a limited sensitivity analysis, not an exhaustive characterization of sampling variability. Because fold membership changes with the selected subset, the present design cannot assign that variability exclusively to artist composition or exclusively to penalty selection.

Prior exposure limits the historical evidence. All 1,206 historical tracks have been used in earlier project work; the outer splits overlap, and the three representations share those tracks. Broad source tags can be wrong or incomplete, artist aliases are not fully resolved, and encoder pretraining overlap cannot be fully audited. Audio

previews and whole-track tags may also disagree. Hash checks establish consistency with stored records, not musical ground truth or the absence of external leakage. The study cannot establish performance on arbitrary recordings or on another data source.

Brier improvement is not a complete calibration claim. The archived reliability bins and label-level results support inspection, but small bin counts limit interpretation. The historical overlapping splits are not treated as independent samples for constructing population confidence intervals. The new evaluation uses artist-level resampling conditional on fixed heads and one calibration pool; it does not remove uncertainty about other calibration populations or data sources. The parameter bounds, penalty grid and implementation choices also limit transfer to other calibrators. Cross-platform success removes one computational ambiguity without resolving these statistical limitations.

The prospective evaluation adds evidence beyond the previously examined tracks: all selected artists are disjoint from the audited historical ledger and the primary comparison follows the frozen rules. The historical ledger contains 2,835 tracks and 1,296 artist IDs, with additional credited-name exclusions used for the new selection. This protection is limited to documented project exposure and the specified exact-audio checks. The new cohort remains part of MTG-Jamendo, uses the same source-tag ontology and shares one calibration pool across its five draws. Its label-independent artist sampling and two-track cap also change the sampling distribution relative to the historical coverage-enriched cohort. The conditional interval cannot remove those limitations. A second source with a separately frozen protocol and carefully matched label meaning would test a different generalization claim.

7 Conclusion

The historical benchmark documents both the benefit of regularization relative to ordinary Platt and the risk of unnecessary calibration relative to raw or analytical controls. The new-artist evaluation supports its prespecified partial-budget mean ridge-over-Platt comparison, with Brier difference -0.007050 and a conditional 95% interval below zero. It does not establish an advantage over the simpler controls, and the full-budget ridge-versus-Platt intervals span zero. Numerical verification makes the comparisons auditable without turning same-source, noisy-label evidence into a universal calibration guarantee. The contribution is a reproducible empirical study with a prospectively specified test, not a new calibration algorithm.

Data and code availability

Historical code, score bundles, frozen protocols and compact verification records are available at <https://github.com/f0rKyrie1rving/music-probability-calibration>, commit 5577baf3c0048827fa5e71a940a08fd22f4a641f. That commit identifies the historical evidence, not the new confirmation artifacts. The numerical implementation was tested in CI at candidate 816dec19044ce45d0496c4f720d33cc419309a52; subsequent historical documentation and evidence do not change those frozen numerical sources. The historical source score bundle records upstream application commit 92bc9fb8090f1c8c18c63b6dcea25ae04030b9de. Full probability signatures can be regenerated; hosted CI artifacts have retention limits.

The tested environment uses Python 3.13.15, NumPy 2.5.2, SciPy 1.18.1 and scikit-learn 1.9.0. Protocols and environment receipts are part of the repository. The software is MIT-licensed. Source-derived score tables and labels are supplied for non-commercial research under CC BY-NC-SA 4.0 with MTG-Jamendo attribution; original audio retains per-recording licenses. Audio, encoder weights and full feature caches are not redistributed in the score bundle.

The paper-v1 release of the repository provides the new confirmation protocol, score-level inputs, fitted calibration records, source attribution and summary results. Its verification script reconstructs predictions and reported comparisons without the original application or audio. The public package documents its

relationship to the original local freeze and execution records; it does not reproduce audio acquisition or feature extraction. Full local execution records and audio are not redistributed. The new run uses the same NumPy, SciPy and scikit-learn versions as the historical verification, with PyTorch 2.8.0 for feature extraction. See the repository README and reports/fresh_confirmation_v1 for the reproduction boundary and commands.

Research assistance and author responsibility

The project was initiated under Chenglin Song’s direction. Generative AI assisted with literature retrieval, experimental design, programming, tests, numerical verification and manuscript drafting and translation. The paper reports computed results from preserved source-derived data; no synthetic observations or invented labels were added. This preprint is released for public discussion and has not undergone peer review. Chenglin Song is responsible for its scientific interpretation, references and any future submission declarations.

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Appendix A Complete treatment comparisons

Tables A1 and A2 show mean macro Brier scores for every treatment. Partial-budget cells use five draw means within each of 20 outer splits. Full-budget cells use one endpoint per split. Lower is better; rounding is for display only. Repeated raw and prior-offset values are retained to make the comparator explicit. U denotes unweighted heads and W denotes ambient-balanced heads where used.

Table A1. Five-draw stability summary for unweighted classifiers. All seven treatments are retained.

Representation	Artists	Raw	Offset	Temp.	Platt	Beta	Isotonic	Ridge
MAEST	24	0.13216	0.13216	0.13580	0.14113	0.14198	0.14830	0.13535
MAEST	47	0.13216	0.13216	0.13383	0.13608	0.13667	0.14110	0.13322
MAEST	94	0.13216	0.13216	0.13281	0.13340	0.13374	0.13712	0.13251
MERT-v0	24	0.14399	0.14399	0.14766	0.15215	0.15234	0.15847	0.14717
MERT-v0	47	0.14399	0.14399	0.14508	0.14671	0.14662	0.15110	0.14457
MERT-v0	94	0.14399	0.14399	0.14403	0.14410	0.14383	0.14708	0.14374
MERT-v1	24	0.14402	0.14402	0.14791	0.15240	0.15262	0.15994	0.14724
MERT-v1	47	0.14402	0.14402	0.14534	0.14682	0.14685	0.15182	0.14475
MERT-v1	94	0.14402	0.14402	0.14425	0.14429	0.14403	0.14696	0.14375

Table A2. Five-draw stability summary for ambient-balanced classifiers. Only the ambient head changes between weighting settings.

Representation	Artists	Raw	Offset	Temp.	Platt	Beta	Isotonic	Ridge
MAEST	24	0.14044	0.13275	0.14413	0.14127	0.14208	0.14855	0.13799
MAEST	47	0.14044	0.13275	0.14222	0.13621	0.13680	0.14147	0.13433
MAEST	94	0.14044	0.13275	0.14121	0.13350	0.13386	0.13735	0.13259
MERT-v0	24	0.15575	0.14422	0.15927	0.15209	0.15202	0.15834	0.14979
MERT-v0	47	0.15575	0.14422	0.15693	0.14664	0.14639	0.15079	0.14571
MERT-v0	94	0.15575	0.14422	0.15586	0.14406	0.14366	0.14697	0.14394
MERT-v1	24	0.15626	0.14420	0.16004	0.15238	0.15243	0.16003	0.14960
MERT-v1	47	0.15626	0.14420	0.15759	0.14678	0.14666	0.15177	0.14591
MERT-v1	94	0.15626	0.14420	0.15652	0.14425	0.14387	0.14709	0.14396

Appendix B Original draw and exact experimental settings

The original budget draw is a historical reference and is not pooled into the five-draw estimates. Table B1 retains its ridge contrasts for every representation, weighting and budget. The full seven-method original-draw tables remain in reports/numerical_v2/budget_original. Outer split seeds are 2026092800 through 2026092819. The original draw seed is 2026100101; follow-up seeds are 2026100201 through 2026100205; the inner fold seed is 2026100102.

Table B1. Original-draw mean paired Brier contrasts for ridge Platt. Each entry averages 20 outer splits; negative values favor ridge.

Representation	Weight	Artists	Difference from raw	Difference from Platt
MAEST	U	24	+0.003745	-0.008067
MAEST	U	47	+0.000601	-0.003623
MAEST	U	94	+0.000351	-0.000885
MAEST	W	24	-0.002395	-0.005957
MAEST	W	47	-0.006106	-0.002154
MAEST	W	94	-0.007849	-0.000908
MERT-v0	U	24	+0.003492	-0.007470
MERT-v0	U	47	+0.000316	-0.003038
MERT-v0	U	94	-0.000252	-0.000355
MERT-v0	W	24	-0.006379	-0.005419
MERT-v0	W	47	-0.009726	-0.001289
MERT-v0	W	94	-0.011809	-0.000120
MERT-v1	U	24	+0.003731	-0.007815
MERT-v1	U	47	+0.000945	-0.002960
MERT-v1	U	94	-0.000270	-0.000543
MERT-v1	W	24	-0.006151	-0.005273
MERT-v1	W	47	-0.009257	-0.000832
MERT-v1	W	94	-0.012297	-0.000293

For numerical reproduction, the solver uses an Armijo constant of 10^{-4} , at most 55 step-halving attempts, a Cholesky pivot floor of 10^{-14} times the largest Hessian diagonal, and at most 80 bisection steps per coordinate. A roundoff-scale loss change is accepted only when the projected-gradient residual decreases. The frozen numerical protocol and independent verification sources specify the remaining arithmetic checks and saved diagnostics. These settings were fixed before the six complete v2 runs.

Appendix C Prospective protocol and complete method comparisons

The new protocol was fixed before new audio acquisition and derives its sample-size target only from historical predictions. Across the 20 historical splits, the 90th percentile of the artist-cluster track-ratio influence standard deviation for the proposed primary contrast was approximately 0.01734. A normal approximation gives a 95% half-width of approximately 0.00170 at 400 artists and 0.00190 at 320. A rounded planning standard deviation of 0.02 gives a half-width of 0.00196 at 400. These are planning approximations, not guarantees of power or precision in the new sampling distribution.

The metadata source is pinned to MTG-Jamendo commit `cafd8e20c265ed84f1e61f1c875327971f43a62f`. The fixed sampling seed is `music-calibration-fresh-confirmation-20261003-v1`. Permitted recording licenses are CC BY, BY-SA, BY-NC and BY-NC-SA. The recorded historical track/artist ledger is augmented with three conservatively excluded artist IDs whose normalized credited names collide with historical names. A name collision is an exclusion precaution, not proof of identity. Downloading uses bounded audio prefixes for the first 30 seconds. A prefix hash is not reported as a verified full-recording checksum. Exact duplicate checks compare canonical 30-second mono 16-kHz PCM16 excerpts and cannot exclude every re-encoding, cover or unknown pretraining occurrence. Audio retains its per-recording license and is not redistributed. The full prospective freeze SHA-256 is preserved in the local freeze.json; its identifier begins `2af3f1b7222d`.

Calibration draw seeds are 2026100301 through 2026100305, the three-fold artist assignment seed is 2026100311, and the evaluation bootstrap seed is 2026100321. Missing rows are intersected with frozen subsets without reassignment or replacement. The 94-artist plans and fold assignments are identical across draws and counted once, giving $20 \times 2 \times (5 \times 2 + 1) = 440$ distinct planned head, weighting and subset conditions. All seven treatments use the existing numerical-v2 implementation. Every optimized candidate, including unselected inner candidates, must converge; source-label single-class folds retain the recorded identity fallback.

The fit stage hashes the extraction outputs, uses only calibration targets for fitting and selection, and saves all fitted maps, candidate out-of-fold records and evaluation predictions before the report stage parses evaluation labels. This auditable separation is not third-party blinded custody: source metadata remained available to the experimenter. Unaffected labels must match exactly between weighting settings, and raw/offset predictions must remain unchanged across subsets. Numerical or source-integrity failures stop the run rather than being discarded. The historical three-platform verification remains separate from any execution checks performed for this new cohort.

Tables C1 and C2 retain every method at each nominal budget and weighting setting. Scores average the losses of fixed head/subset conditions, with full-budget repetitions counted once. Actual calibration counts and attrition are reported in Table 4. Label-level Brier, average precision, reliability bins, fit diagnostics and selections remain in the corresponding run records. The unweighted analytical offset equals raw exactly and is retained as a transparent control.

Table C1. Complete new-cohort mean macro Brier scores. The three budgets are nominal artist counts; actual counts follow the frozen attrition rules. U denotes unweighted heads and W denotes ambient-balanced heads.

Weight	Artists	Raw	Offset	Temp.	Platt	Beta	Isotonic	Ridge
U	24	0.12503	0.12503	0.12824	0.13956	0.14133	0.14932	0.12876
U	47	0.12503	0.12503	0.12628	0.12867	0.13060	0.13785	0.12537
U	94	0.12503	0.12503	0.12510	0.12440	0.12571	0.13076	0.12476
W	24	0.13674	0.12482	0.14045	0.13951	0.14119	0.14921	0.13126
W	47	0.13674	0.12482	0.13821	0.12874	0.13047	0.13794	0.12614
W	94	0.13674	0.12482	0.13686	0.12449	0.12590	0.13103	0.12462

Table C2. Complete new-cohort mean macro log loss, using the same conditions as Table C1. Probabilities are clipped to the fixed scoring bounds only when calculating log loss.

Weight	Artists	Raw	Offset	Temp.	Platt	Beta	Isotonic	Ridge
U	24	0.39718	0.39718	0.41894	0.50371	0.54382	1.98488	0.42354
U	47	0.39718	0.39718	0.40360	0.41056	0.42244	1.09967	0.39840
U	94	0.39718	0.39718	0.39666	0.39504	0.39834	0.51486	0.39646
W	24	0.42715	0.39682	0.44904	0.50461	0.54842	1.97093	0.43069
W	47	0.42715	0.39682	0.43340	0.41074	0.42221	1.09387	0.40060
W	94	0.42715	0.39682	0.42676	0.39529	0.39891	0.51674	0.39627